A Lightweight CNN Approach for Hand Gesture Recognition via GAF Encoding of A-Mode Ultrasound Signals

Qican Shangguan, Yue Lian, Zhiwei Liao, Jinshui Chen, Yiru Song, Ligang Yao, Cai Jiang, Zongxing Lu, *Member, IEEE*, and Zhonghua Lin

***Abstract*— Hand gesture recognition(HGR) is a key technology in human-computer interaction and human communication. This paper presents a lightweight, parameter-free attention convolutional neural network (LPA-CNN) approach leveraging Gramian Angular Field(GAF)transformation of A-mode ultrasound signals for HGR. First, this paper maps 1-dimensional (1D) A-mode ultrasound signals, collected from the forearm muscles of 10 healthy participants, into 2-dimensional (2D) images. Second, GAF is selected owing to its higher sensitivity against Markov Transition Field (MTF) and Recurrence Plot (RP) in HGR. Third, a novel LPA-CNN consisting of four components, i.e., a convolution-pooling block, an attention mechanism, an inverted residual block, and a classification block, is proposed. Among them, the convolution-pooling block consists of convolutional and pooling layers, the attention mechanism is applied to generate 3-D weights, the inverted residual block consists of multiple channel shuffling units, and the classification block is performed through fully connected layers. Fourth, comparative experiments were conducted on GoogLeNet, MobileNet, and LPA-CNN to validate the effectiveness of the proposed method. Experimental results show that compared to GoogLeNet and MobileNet, LPA-CNN has a smaller model size and better recognition performance, achieving a classification accuracy of 0.98 ±0.02. This paper achieves efficient and high-accuracy HGR by encoding A-mode ultrasound signals into 2D images and integrating the LPA-CNN model, providing a new technological approach for HGR based on ultrasonic signals.**

***Index Terms*— A-mode ultrasound, convolutional neural network (CNN), deep learning, hand gesture recognition (HGR), gramian angular field (GAF).**

I. Introduction

With the development of technology and the continuous increase in computing power, human-computer interaction has become a key research focus. Among the various aspects of human body language, gestures stand out as one of the most important, offering the advantages of being concise, intuitive, and information-rich [1], [2]. As a result, hand gesture recognition (HGR) technology has attracted significant attention. Currently, the main hand gesture sensing methods include data gloves [3], [4], [5], vision-based methods [6], [7], [8], surface electromyography (sEMG) [9], [10], [11], [12], and ultrasound [13], [14], [15]. Fan et al. [16] utilized data gloves to capture finger data and applied the SqueezeNet-BiLSTM algorithm for recognition, achieving an accuracy of 98.21%. Sharma and Singh [17] used convolutional neural network (CNN) to recognize vision-based hand gesture data, yielding excellent classification results. However, these sensors have some limitations [18], [19], [20]: data gloves are inconvenient to wear and offer a poor user experience; vision sensors are highly affected by lighting conditions and offer poor privacy; sEMG signals are weak, exhibit a poor signal-to-noise ratio, and are prone to interference. Researchers have found that A-mode ultrasound has advantages such as portability, high signal-to-noise ratio, and good stability, making it suitable for HGR research [21], [22], [23], [24].

HGR fundamentally involves classifying labeled data samples using supervised machine learning algorithms. Due to improvements in GPU algorithms, deep learning has become increasingly popular in classification tasks, leading many researchers to explore the integration of bio-signals with deep learning. For example, Tan et al. [25] employed a CNN to conduct experiments on multiple gesture datasets, achieving

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an average accuracy of 98.5%; Lian et al. [26] proposed a Squeeze-Excitation CNN for HGR and applied weight fine-tuning among individuals to achieve high precision. From these studies, it can be seen that using deep learning models to improve HGR accuracy and enhance the cross-individual generalization ability of models is the mainstream research. However, there is a scarcity of research on model lightweighting.

A good representation of time-series features not only aligns well with the model and reduces computational complexity but also improves recognition accuracy, which has led many researchers to devote significant effort to this area [27], [28], [29]. A highly effective approach involves representing features as visual cues [30], [31]. By utilizing both supervised and unsupervised learning methods in computer vision, time-series data can be re-encoded into images, allowing machines to carry out image recognition tasks. This technique has been successfully applied to areas such as speech recognition [32], classification [33], and radio frequency identification [34], demonstrating its high effectiveness.

Therefore, this paper extracts more representative features of A-mode ultrasound signals through various comparative experiments, simplifying the complexity of the signals and inverse mapping the characteristic signals of gestures into an encoded image dataset. Additionally, a lightweight convolutional neural network framework is designed for the gesture encoded images, significantly reducing the model's training time while achieving high-precision recognition results.

The rest of this paper is organized as follows: Section II presents the theory and methodology, Section III describes the experimental equipment and design, the experimental result are statistically presented in Section IV, the discussion is presented in Section V, and Section VI concludes and predicts the future work.

II. Methods

A. Ultrasonic Theory

The human forearm contains muscles such as the extensor digitorum longus(EDL), flexor digitorum profundus(FDP), extensor pollicis longus(EPL) and flexor pollicis longus(FPL), which control different hand gestures [35], [36]. By placing an A-mode ultrasound probe on the surface of these forearm muscles, as shown in Fig. 1, echo signals containing information about the morphological changes in the muscles can be collected. Based on information such as the amplitude of the waveform and the return time of the echo signal, it is possible to determine the type of tissue, location, and structural morphology of the forearm muscles.

B. Graphical Encoding

The Gramian Angular Field (GAF) is an image encoding method that preserves the temporal information of 1D signals using angular summation and difference. Given a time series with n data points $X = \{x_1\ x_2 \cdots x_n\}$, the time series is normalized so that all data points fall within the interval [0, 1]. The feature quantization formula is as follows:

$$\bar{x}_i = \frac{x_i - \min(X)}{\max(X) - \min(X)} \tag{1}$$

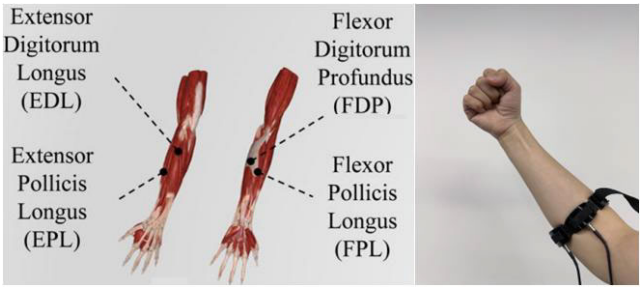


Fig. 1. The main muscles in the forearm that control gestures, and the ultrasound probe fixation method and position.

After normalization, the time series data can be transformed using a polar coordinate system. The values of the data points are mapped to angles through a cosine function, while the time points are represented by the radius. The transformation is defined as follows:

$$\begin{cases} \phi = \arccos(\bar{x}_i) \\ r = \dfrac{t_i}{N} \end{cases} \tag{2}$$

where t_i is the timestamp, and N is a fixed scalar factor representing the normalized radial distance in the polar coordinate system. This representation based on polar coordinates provides a way to interpret time series data. According to the formula above, this encoding has two key features. First, it guarantees a one-to-one mapping due to the monotonicity of the mapping function under certain conditions, ensuring that the polar coordinate mapping of any given time series produces a unique result with a unique inverse mapping. Second, unlike the Cartesian coordinate system, the polar coordinate system preserves the absolute distance relationships between time points.

After converting to the polar coordinate system, analyzing angular relationships becomes convenient. By analyzing trigonometric relationships, such as summation and difference between points, it helps capture time series correlations over different periods. The method of analyzing the angles between points is referred to as the Gramian Angular Summation Field (GASF), while the method of analyzing the angular differences between points is called the Gramian Angular Difference Field (GADF). The GASF and GADF are defined in the following manner:

$$GASF = \begin{bmatrix} \cos(\phi_1 + \phi_1) & \cdots & \cos(\phi_1 + \phi_n) \\ \cos(\phi_2 + \phi_1) & \cdots & \cos(\phi_2 + \phi_n) \\ \vdots & \ddots & \vdots \\ \cos(\phi_n + \phi_1) & \cdots & \cos(\phi_n + \phi_n) \end{bmatrix} \tag{3}$$

$$= \bar{X}' \cdot \bar{X} - \sqrt{I - \bar{X}^2}' \cdot \sqrt{I - \bar{X}^2} \tag{4}$$

$$GADF = \begin{bmatrix} \sin(\phi_1 - \phi_1) & \cdots & \sin(\phi_1 - \phi_n) \\ \sin(\phi_2 - \phi_1) & \cdots & \sin(\phi_2 - \phi_n) \\ \vdots & \ddots & \vdots \\ \sin(\phi_n - \phi_1) & \cdots & \sin(\phi_n - \phi_n) \end{bmatrix} \tag{5}$$

$$= \sqrt{I - \bar{X}^2}' \cdot \bar{X} - \bar{X}' \cdot \sqrt{I - \bar{X}^2} \tag{6}$$

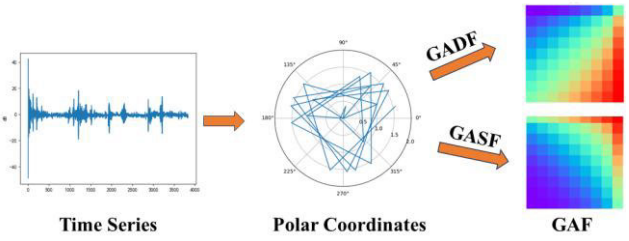


Fig. 2. Depiction of the suggested encoding map for the gramian angular field.

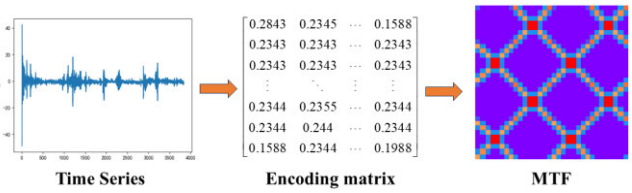


Fig. 3. Depiction of the suggested encoding map for the Markov transition fields.

where I is a unit vector. After converting the time series into a polar coordinate system, each moment in the time series is treated as an independent dimension. The two Gramian angular fields, by defining the inner product $\langle x, y\rangle = x\cdot y - \sqrt{1-x^2}\cdot\sqrt{1-y^2}$ and $\langle x, y\rangle = \sqrt{1-x^2}\cdot y - x\cdot\sqrt{1-y^2}$, essentially form a Gramian matrix:

$$G = \begin{bmatrix} \langle \bar{x}_1, \bar{x}_2\rangle & \cdots & \langle \bar{x}_1, \bar{x}_n\rangle \\ \langle \bar{x}_2, \bar{x}_1\rangle & \cdots & \langle \bar{x}_2, \bar{x}_n\rangle \\ \vdots & \ddots & \vdots \\ \langle \bar{x}_n, \bar{x}_1\rangle & \cdots & \langle \bar{x}_n, \bar{x}_n\rangle \end{bmatrix} \tag{7}$$

Since$G_{(i,j||i-j|=k)}$ reflects the relative dependency of the superposition (or difference) of direction on the time interval k, this characteristic enables the GAF image to capture time dependencies. Additionally, the main diagonal represents the special case when $k = 0$, containing the original angular information [37]. By utilizing the encoding matrix, the time series can be reconstructed into a GAF image, and its generation process is shown in Fig. 2.

The Markov Transition Field (MTF) is another image encoding method that preserves temporal information by representing Markov transition probabilities. The encoding matrix is defined as follows:

$$\begin{cases} M = \begin{bmatrix} w_{ij|x_1\in q_i, x_1\in q_j} & \cdots & w_{ij|x_1\in q_i, x_n\in q_j} \\ \vdots & \ddots & \vdots \\ w_{ij|x_n\in q_i, x_1\in q_j} & \cdots & w_{ij|x_n\in q_i, x_n\in q_j} \end{bmatrix} \\ w_{i,j} = \sum_{\forall x\in q_i, y\in q_j, x+1=y} 1/\sum_{j=1}^{Q} w_{i,j} \end{cases} \tag{8}$$

where Q is the total number of quantile bins, $q_i$and q_j are the quantile bins that encompass data at time stamp i and j, w_{ij}in the MTF denotes the transition probability of $q_i \to q_j$.

The encoding matrix can be used to generate a corresponding MTF image using Python, and the generation process is illustrated in Fig 3. The Recurrence Plot (RP) is a technique

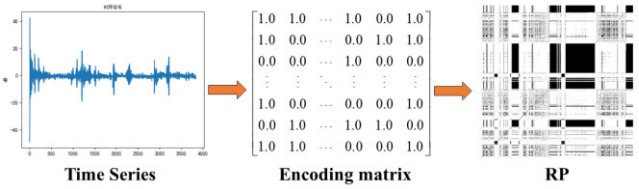


Fig. 4. Depiction of the suggested encoding map for the recurrence Plot.

for transforming time series data into visual images by representing the similarity between points in the time series. Its generation process is shown in Fig.4. This method requires reconstructing the time series, defined as follows:

$$\bar{X} = \left\{x_i, x_{i+\tau}, \cdots, x_{i+(m-1)\tau}\right\} \tag{9}$$

where $\tau$is the time delay and $m$is the embedding dimension. Then, the encoding matrix is calculated from the reconstructed time series, defined as:

$$\begin{cases} R = \theta(\varepsilon - \left\|\bar{X}_i - \bar{X}_j\right\|) \\ \theta(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases} \end{cases} \tag{10}$$

where ε is a threshold and $\|\cdot\|$is a function that calculates the Euclidean distance.

C. LPA-CNN

Traditional CNN such as AlexNet, VGG, and GoogLeNet, although having strong classification abilities [38], [39], [40], often suffer from long training times and high computational costs. In response to this, we propose a lightweight, parameter-free attention convolutional neural network (LPA-CNN). The framework of the model is shown in Fig. 5. The backbone network of LPA-CNN consists of four main components:

1) CP Block: The convolution-pooling block (CP Block) is composed of a convolutional layer and a pooling layer. The convolutional layer uses a fixed-size kernel and stride to filter the input data. It also contains a batch normalization(BN) layer to accelerate convergence and prevent overfitting, while the ReLU activation function helps speed up training. The input and output data dimensions are as follows:

$$\begin{cases} OH = \dfrac{H + 2p - k}{s} + 1 \\ OW = \dfrac{W + 2p - k}{s} + 1 \end{cases} \tag{11}$$

where k is the convolution kernel size, p is the padding value, s is the step size. The pooling layer follows closely, applying max pooling to the output of the convolutional layer, which extracts the maximum value from each window. This process reduces the dimensionality of the data and decreases the size of the parameter matrix, improving the computational efficiency of the model.

2) SimAM Block: The SimAM Block denotes the simple, parameter-free attention mechanism. Inspired by well-known neuroscience theories, this mechanism optimizes an energy function to determine the importance of each neuron [18].

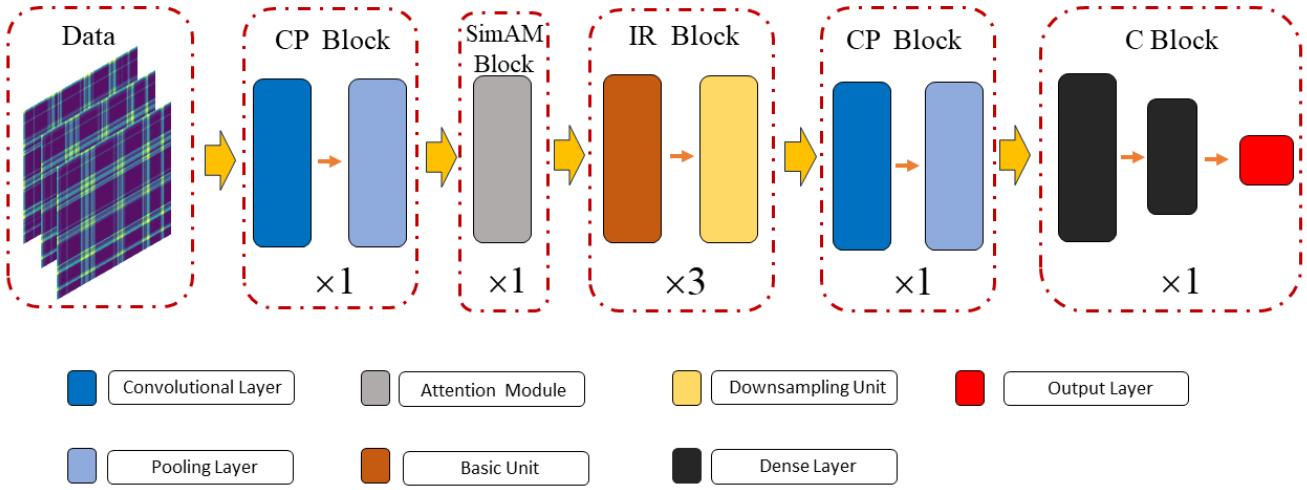


Fig. 5. The lightweight, parameter-free attention convolutional neural network mode. 1)The CP Block is a traditional block composed of a convolutional layer and a pooling layer. 2) The SimAM Block denotes the simple, parameter-free attention mechanism. 3) The inverted residual Block (IR Block) is composed of multiple channel shuffle units. 4) C Block stands for classification block.

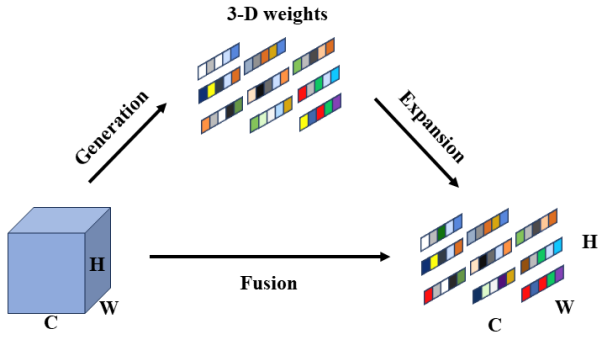


Fig. 6. The simple, parameter-free attention mechanism.

The energy function is defined as the sum of the similarities between a neuron and its neighboring neurons, with the specific form given as:

$$E = \frac{1}{M-1}\sum_{i=1}^{M-1}(-1-(w_t x_i + b_t))^2 + (1-(w_t t + b_t))^2 + \lambda w_t^2 \tag{12}$$

where M is the number of neurons in the channel, w_t and b_t are the weights and biases of the transformation, λ is the regularization coefficient, t is the target neuron, x_i represents other neurons in the same channel.

As shown in Fig. 6, the 3-D attention weights for each neuron can be computed by optimizing the above energy function. These weights are then used to perform the Expansion step, where they are applied to the feature map through element-wise multiplication, enhancing important features while suppressing less relevant ones. Finally, the Fusion step is carried out to integrate the weighted feature maps, combining information from multiple dimensions. Since the SimAM Block does not introduce any additional parameters, it not only keeps the model lightweight but also improves feature robustness and accuracy.

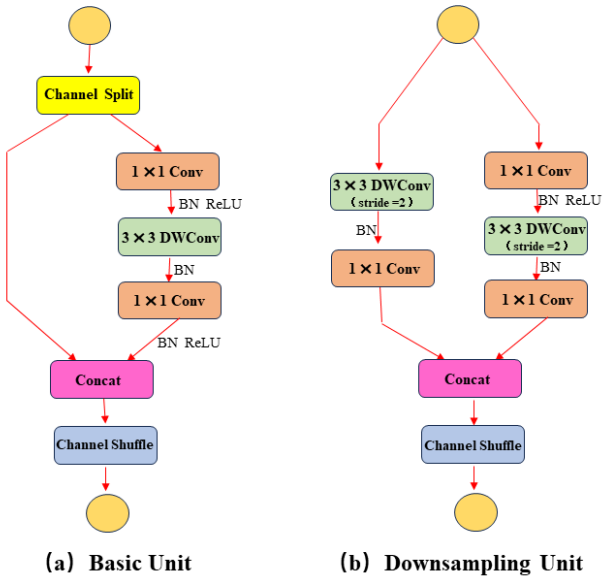


Fig. 7. (a) The basic unit; (b) The unit for spatial down sampling.

3) IR Block: The inverted residual block (IR Block) is composed of multiple channel shuffle units. Fig. 7 illustrates two types of channel shuffle units. In the basic unit, feature channels are divided to reduce the computational parameters, while the spatial down-sampling unit uses two convolution branches to double the output channels, enhancing the network's expressive power. This block also introduces channel shuffling to ensure information exchange between branches, avoiding information loss [41]. This mechanism enables the network to deeply mine data features, thereby improving model accuracy.

4) C Block: The classification block (C Block) is made up of two fully connected layers with dimensions of 1024 and 256, respectively, utilizing the ReLU activation function. The output is set to 10 units to align with the ten hand gesture labels

designed for this study. This gradual dimensionality reduction helps prevent overfitting and improves model performance.

Additionally, the LPA-CNN uses the Adam optimizer with a learning set at 0.001 and utilizes the Cross-Entropy loss function to assess the error between true and predicted labels. Adam adjusts the learning rate dynamically for different parameters based on the first and second moments of the gradients. Its bias correction ensures smoother parameter updates. Cross-entropy, commonly used for classification tasks, measures the distribution distance between two probabilities, helping the model better approximate the actual results. It is defined as:

$$L = -\frac{1}{N}\sum_{i}\sum_{c=1}^{M} y_{ic}\log(p_{ic}) \tag{13}$$

where M is the number of categories, y_{ic} is 1 when the true category of sample i is equal to c and otherwise,and p is the predicted probability that sample i belongs to category c.

D. Evaluation Method

To evaluate the results of our experiments, we have employed several metrics, including the Silhouette Coefficient (SC), recognition accuracy (RA), Floating Point Operations (FLOPs), and classification error rate (CER).

SC is a measure used to assess the quality of clustering outcomes. An SC value close to 1 indicates a clearer silhouette and a more distinct clustering effect. Its definition is as follows:

$$\begin{cases} s_i = \dfrac{b_i - a_i}{\max(a_i, b_i)} \\ S = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N} s_i \end{cases} \tag{14}$$

where a_i represents the average distance between a point and all other points within the same cluster, and b_i denotes the minimum average distance of i from the samples in the other clusters.

RA is one of the most important metrics for evaluating the quality of recognition results. The closer the RA value is to 1, the better the recognition performance. The formula is as follows:

$$RA = \frac{N_{right}}{N_{all}} \times 100\% \tag{15}$$

FLOPs is a key metric used to measure the computational cost and complexity of a model. The smaller the FLOPs, the more lightweight the model, indicating higher efficiency. Its definition is as follows:

$$FLOPs = 2 \cdot K_H \cdot K_W \cdot C_{in} \cdot C_{out} \cdot O_H \cdot O_W + 2 \cdot N_{in} \cdot N_{out} \tag{16}$$

where K_H and K_W are the height and width of the convolution kernel, C_{in} and C_{out} are the numbers of input and output channels, O_H and O_W are the height and width of the output feature map, N_{in} and N_{out} are the numbers of input and output neurons.

In addition, to highlight the differences in results among different models, we first conducted the Kolmogorov-Smirnov (KS) test to assess the normality of the data. Upon satisfying the normality assumption, the analysis of variance (ANOVA) was performed to detect overall differences. To further identify specific model pairs with significant differences, Tukey's Honestly Significant Difference (Tukey HSD) test was applied for post hoc pairwise comparisons. All statistical analyses were conducted at a significance level of 0.05.

III. Experiment

A. Materials

This study designs two types of hand gestures: 1) natural hand gestures, including clenched fist (CF), stretched thumb (ST), stretched index finger (SI), stretched thumb and index finger (STI), stretched index and middle finger (STM); and 2) grasping hand gestures, including cylindrical grasp (CG), spherical grasp (SG), lateral pinch grasp (LPG), pinch grasp(PG), fingertip grasp (FG). These ten experimental gestures are shown in Fig. 8(a). This experiment uses the A-mode ultrasound device manufactured by ELONXI. The sampling frequency of this ultrasound device is 20 MHz, the operating frequency of the probe is 2.25 MHz, and each probe collects 1000 sampling points. Its timer acquires 10 frames of data per second. During the experiment, four A-mode ultrasound probes were placed on four forearm muscles: EDL, EPL, FDP and FPL. Ultrasound coupling gel was applied between the probes and the skin to minimize signal loss.

The study protocol was approved by the Ethics Committee of Affiliated Provincial Hospital of Fuzhou University (No: K2025-03-019), and all participants provided written informed consent after receiving education on the study's purpose and procedures. The participants included 10 subjects aged between 23 and 26, with 7 male and 3 female participants. As illustrated in Fig. 8(b), each subject was guided by a custom GUI and performed 10 gestures in a sequence, with each gesture lasting for 5 seconds. One complete round of gesture collection consists of all ten gestures, and five rounds were performed for each subject. To prevent muscle fatigue, subjects took 3 to 5-minute breaks between rounds.

B. Encoding Preprocessing

When the raw time series has a length of n, it follows from Equation (9) that the dimensions of the encoding matrix are $n \times n$, which increases memory usage and raises the training cost of the CNN. Therefore, it is necessary to perform preprocessing on the collected A-mode ultrasound data. This preprocessing helps recover the true signal from the ultrasound data, removes probe-induced noise, reduces dimensionality, and optimizes the encoding image.

The first step in the encoding preprocessing is the A-mode signal data preprocessing, starting with Time Gain Compensation (TGC), which compensates for the signal attenuation as the ultrasound signal travels further. The TGC is defined as:

$$T(t) = x(t)e^{\alpha f c t} \tag{17}$$

where $x(t)$ and $T(t)$ denote the ultrasound signals prior to and following TGC, α is the attenuation coefficient, f is the

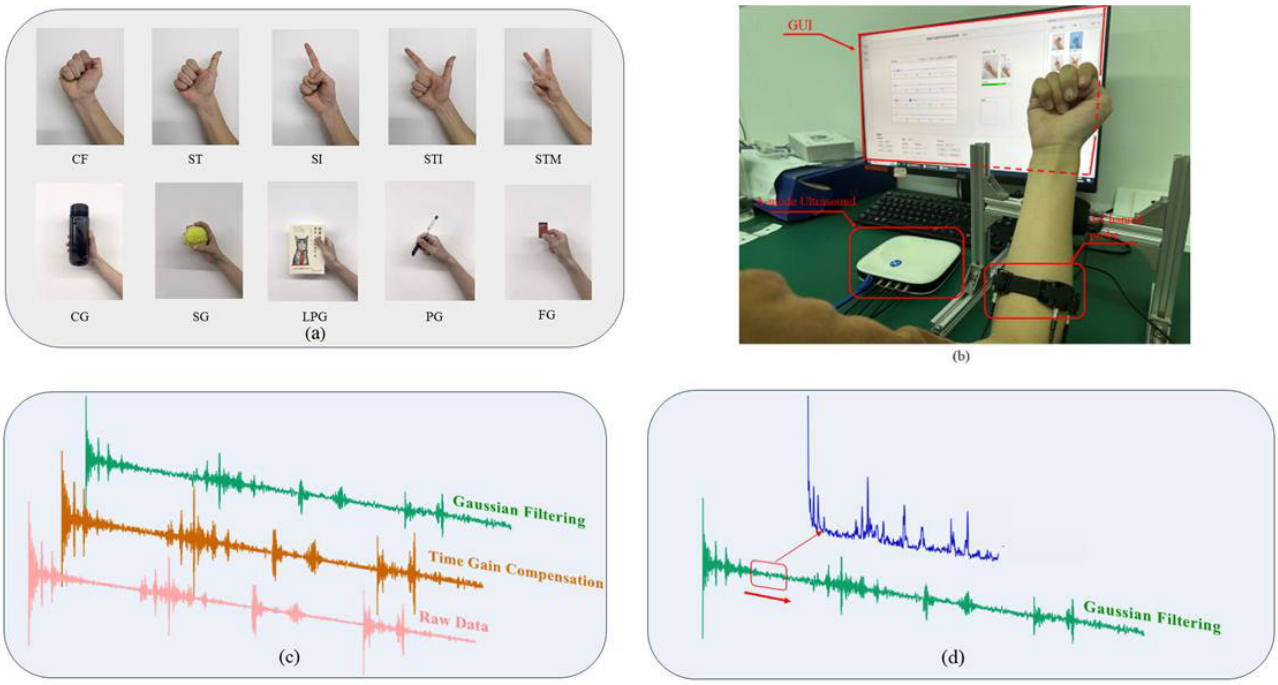


Fig. 8. Experimental equipment and related materials for A-mode ultrasound HGR. (a) Pictures of ten gestures. (b) Experiment full view. (c) Preprocessing process. (d) Feature extraction.

TABLE I
FEATURE EXTRACTION METHODS

Feature	Equation
Energy	$d_i = \left(\sum_{\varepsilon=m(i-1)+1}^{m(i-1)+w} g_\varepsilon{}^2 \right)^{\frac{1}{2}}$
Mean	$d_i = \frac{1}{w} \sum_{\varepsilon=w(i-1)+1}^{wi} g_\varepsilon$
Maximum	$d_i = \max\left\{ g_{w(i-1)+1}, g_{w(i-1)+2}, \cdots, g_{wi} \right\}$

ultrasound frequency, c is the speed of sound in the tissue. In this study $\alpha = 0.05db/MHz \cdot cm$, $c = 1540m/s$.

Following TGC, Gaussian filtering is utilized to eliminate noise, thereby smoothing the ultrasound signal. Gaussian filtering is defined as:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \tag{18}$$

where σ is the standard deviation, μ is the mean, and $\sigma = 1$, $\mu = 0$ in this experiment. The first step is shown in Fig. 8(c).

Next, feature extraction is performed to reduce the size of the encoded map. Specifically, representative information of the ultrasonic signal is extracted using a fixed window length and stride. We set the window length to 10 sampling points and the stride to 5, then calculate the feature energy, mean, and maximum value for each window, as illustrated in Table I.

Even though subjects repeated the same gestures across different rounds, fluctuations in the energy peak values of the ultrasound signals occurred. However, the peak locations remained consistent. These fluctuations can reduce the accuracy of the CNN. To address this, we normalize the signal using a Sigmoid function and a threshold b, which maintains the correlation between the peak location and gesture. The normalized signal is defined as:

$$s_i = (1 + e^{-(d_i - b)})^{-1} \tag{19}$$

In this experiment, we empirically set $b = 8$, and the feature extraction process is shown in Fig. 8(d).

C. Experiment Setup

In this study, the deep learning framework was built using Pytorch, and experiments were conducted on a DELL G15 system equipped with an Intel(R) Core(TM) i5-10500H CPU @ 2.50GHz, 16GB RAM, and an NVIDIA RTX3060 GPU. During the experimental process, we used the first four rounds as the training set, the final round as the test set, and half of the test set for validation. The validation set is primarily for model selection and adjustment. Additionally, to avoid randomness in the model training process, the encoded map dataset for each subject undergoes five rounds of training and testing.

Three experiments were designed. In Experiment 1, to verify the effectiveness of the energy signal and Sigmoid function normalization encoding preprocessing method (ES method), we set up two experimental control groups. The first group uses the ES method, while the other two groups use the mean value with Sigmoid function normalization (MS method) and the peak value with Sigmoid function normalization (PS method), respectively. The overall encoding preprocessing

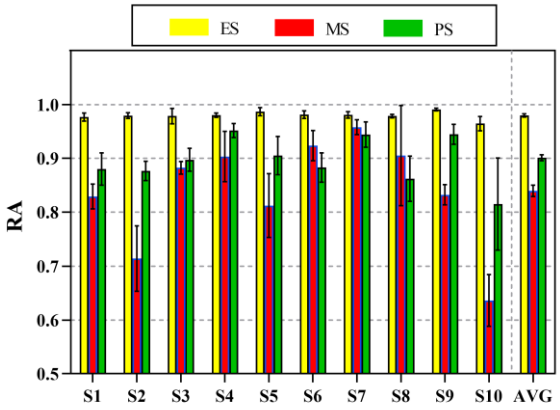


Fig. 9. RA of different encoding preprocessing methods.

procedure for these control groups is the same as for the first group, except for the change in calculation methods during the feature extraction process.

In Experiment 2, to verify the superiority of GAF for classification, we set up two control groups using MTF and RP encoding images. All three types of encoded images used the ES method for preprocessing, as verified in Experiment 1.

In Experiment 3, we built subject-specific models for all 10 participants. To evaluate the performance of our LPA-CNN model in terms of classification and efficiency, we compared it with two control groups: the traditional GoogLeNet model and a lightweight MobileNet model. The GoogLeNet model achieved outstanding results in the ImageNet Large Scale Visual Recognition Challenge and has been widely used in image classification tasks due to its excellent performance. Meanwhile, the MobileNet model is a well-known lightweight model that stands out for its small number of parameters, high recognition accuracy, and efficient inference. Its performance has been proven in multiple studies to surpass other lightweight models(such as SqueezeNet) [42], [43]. Therefore, these two models represent typical high-performance and lightweight models, making them suitable as control groups for comparative analysis. Moreover, to ensure the rigor of the experiment, the three models differ only in their frameworks, while parameters such as batch size, number of training epochs, and learning rate remain the same.

IV. Result and Evaluation

A. Result of Encoding Preprocessing

In Experiment 1, we used SC and RA as evaluation metrics. For each subject's experimental data, the process was repeated five times to obtain five experimental results, and the average of these results was taken as the final metric, as shown in Fig. 9. From Fig. 9, it can be seen that the ES method achieved an accuracy of 98.0%, while the MS and PS methods had accuracies of 93.5% and 94.1%, respectively. Compared to the other two feature processing methods, the ES-based feature extraction method attained the highest RA and exhibited the smallest fluctuation in accuracy across multiple training sessions, demonstrating the best stability.

To make the evaluation results more apparent, we used PCA to reduce the extracted features to 40 dimensions and selected

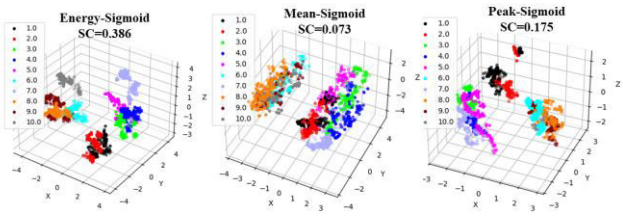


Fig. 10. PCA plots for different encoding preprocessing methods.

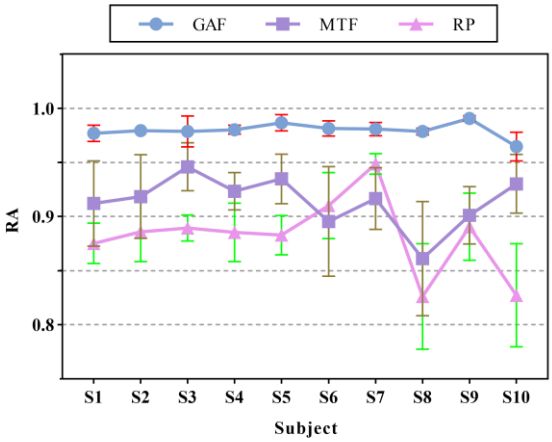


Fig. 11. RA of different encoding images.

the first three dimensions for visualization. In the Cartesian coordinate system, the three axes correspond to the sizes of these three dimensions, with different gestures indicated by dots of various colors. This allows us to roughly observe the distinguishability of different gestures. Fig. 10 shows the PCA results of the three feature extraction methods and their corresponding SC. In the ES method, dots of the same color are more clustered, and the contours between clusters of different colors are clearly distinguishable, indicating better classification performance in the feature space. Conversely, the MS and PS methods exhibit less clustering of dots with the same color and overlapping clusters of different colors, which directly indicates the lower recognition rates.

B. Comparison of Three Encoded Maps

In Experiment 2, we used RA as the evaluation metric. Fig. 11 shows the accuracy of the three encoding images, with the average accuracy of the ten subjects being 98.0% for GAF, 91.3% for MTF, and 88.2% for RP. Fig. 12 provides a more detailed view of the recognition accuracy for the ten gestures.

The recognition accuracy of the GAF encoding image was higher for all gestures compared to the other two encoding images, with only the LPG gesture having slightly lower accuracy than the other gestures, indicating excellent recognition stability. In contrast, the recognition performance of the MTF and RP encoding images for the ten gestures was inconsistent, with significant fluctuations in recognition accuracy between gestures. Notably, the LPG gesture had the lowest recognition accuracy for both the MTF and RP encoding images, at only 0.83 (MTF) and 0.82 (RP).

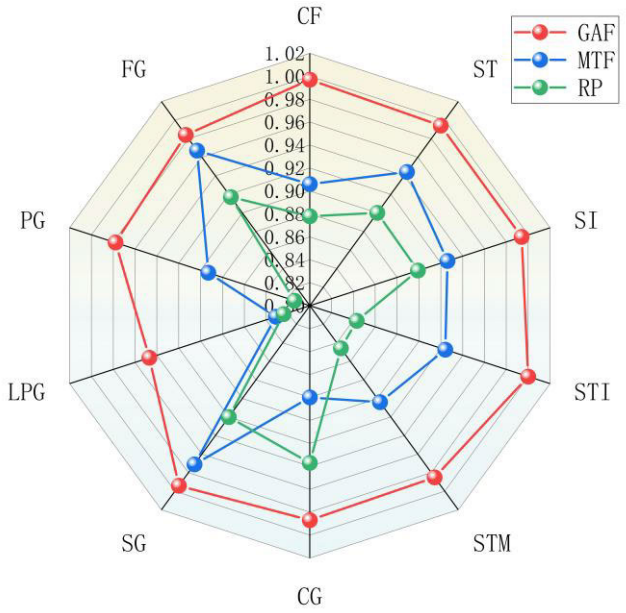


Fig. 12. RA of ten gestures on three types of encoding maps.

C. LPA-CNN's Recognition Performance

In Experiment 3, we used training time, FLOPs, and CER as evaluation metrics. As shown in Fig.13(a), we calculated the FLOPs of the three models using a certain library and found that the FLOPs of the proposed model were only 82 million, significantly lower than the FLOPs of GoogLeNet (2.06 billion) and MobileNet (569 million). A smaller number of FLOPs often corresponds to lower time costs for the model. As shown in Fig. 13(a), under the same number of training rounds, the LPA-CNN model had a training time of 619.33 seconds, saving about 12% of the training time compared to GoogLeNet (704.64 seconds) and MobileNet (690.11 seconds), indicating that our model is a lightweight network framework.

As shown in Fig.13(b), the KS test was conducted on the results of the three recognition models, yielding p-values of 0.1 for LPA-CNN, 0.06 for GoogLeNet, and 0.09 for MobileNet, all of which are greater than the significance level of 0.05. This indicates that, at the current significance level, the recognition result data conforms to the assumption of normal distribution. Fig.13(b) also shows the classification error rates for the three models on the GAF encoding images, and the LPA - CNN model has the lowest average error rate (0.02). Through ANOVA, there are significant differences in the performances of the three models($P<0.001$); further, via the Tukey HSD post hoc test, it is found that the performance of the LPA - CNN model is significantly better than that of GoogLeNet ($P < 0.001$) and MobileNet ($P < 0.05$). Fig. 13(c) provides a more detailed view of the confusion matrix for the ten gestures. Among them, the recognition accuracy of the PG gesture on GoogLeNet is only 0.76, which is a significant drop compared to the recognition result of the PG gesture on LPA-CNN (0.98). Moreover, the PG gesture is primarily misclassified as the LPG gesture. Additionally, the recognition accuracy of the STI gesture and the LPG gesture on MobileNet is only 0.9, which is the worst among the ten gestures. In contrast, the LPA-CNN network model was highly sensitive to all ten gestures, effectively recognizing all of them, with the recognition accuracy for each gesture being higher than that of the other two models.

Fig.14 shows the learning curves of the LPA-CNN model during training, including the changes in training loss and validation loss over the number of epochs. From Fig.14, it can be observed that as the number of epochs increases, both the training loss and validation loss tend to stabilize, and the gap between them remains small. This indicates that the model achieves good fitting performance on both the training and validation sets without signs of overfitting. Specifically, within the first 30 epochs, the training loss rapidly decreases from around 3 to 0.002, while the validation loss drops quickly from around 5 to approximately 0.1. After around 50 epochs, both training and validation losses become stable, maintaining low values, with the validation loss slightly higher than the training loss but stabilized around 0.1. These results demonstrate that the model effectively learns the features of the signals during training.

V. Discussion

This study aims to encode A-mode ultrasound signals into two-dimensional images and develop the LPA-CNN model to achieve efficient and high-accuracy HGR, providing a novel technical solution for ultrasound-based HGR. As shown in Fig.10, the ES feature method exhibits better recognition performance and greater robustness against noise interference and local anomalies than the MS and PS methods. The main reason is that the ES feature extraction method considers the overall distribution of the A-mode ultrasound signal, enabling a more comprehensive representation of its characteristics and the MS and PS methods focus on local extrema of the A-mode signal, potentially overlooking important information and thus compromising recognition performance.

It is worth noting that transforming 1D A-mode ultrasound signals into 2D encoded images not only preserves the temporal information of muscle signals but also helps uncover latent patterns within the data, leading to improved model's recognition capability, as demonstrated by the GAF results in Fig. 11. Among the three types of encoding images used in this study, RP yielded the lowest recognition accuracy due to the binary nature of its encoding matrices, i.e., all values are either zero or one. This binary representation might cause loss of distinctive features during inverse mapping of the A-mode signals, resulting in high similarity across different gestures and poor classification performance. Additionally, the recognition accuracy of GAF-encoded images is higher than that of MTF-encoded images, which can be attributed to the lower uncertainty in the inverse mapping function of GAF. This helps retain more key information during the signal-to-image reconstruction process.

In Experiment 3, the proposed LPA-CNN model exhibited excellent performance and efficiency, which can be primarily attributed to the integration of the SimAM block and IR block. The SimAM block effectively focuses on key information regions, while the IR block enables deeper feature mining

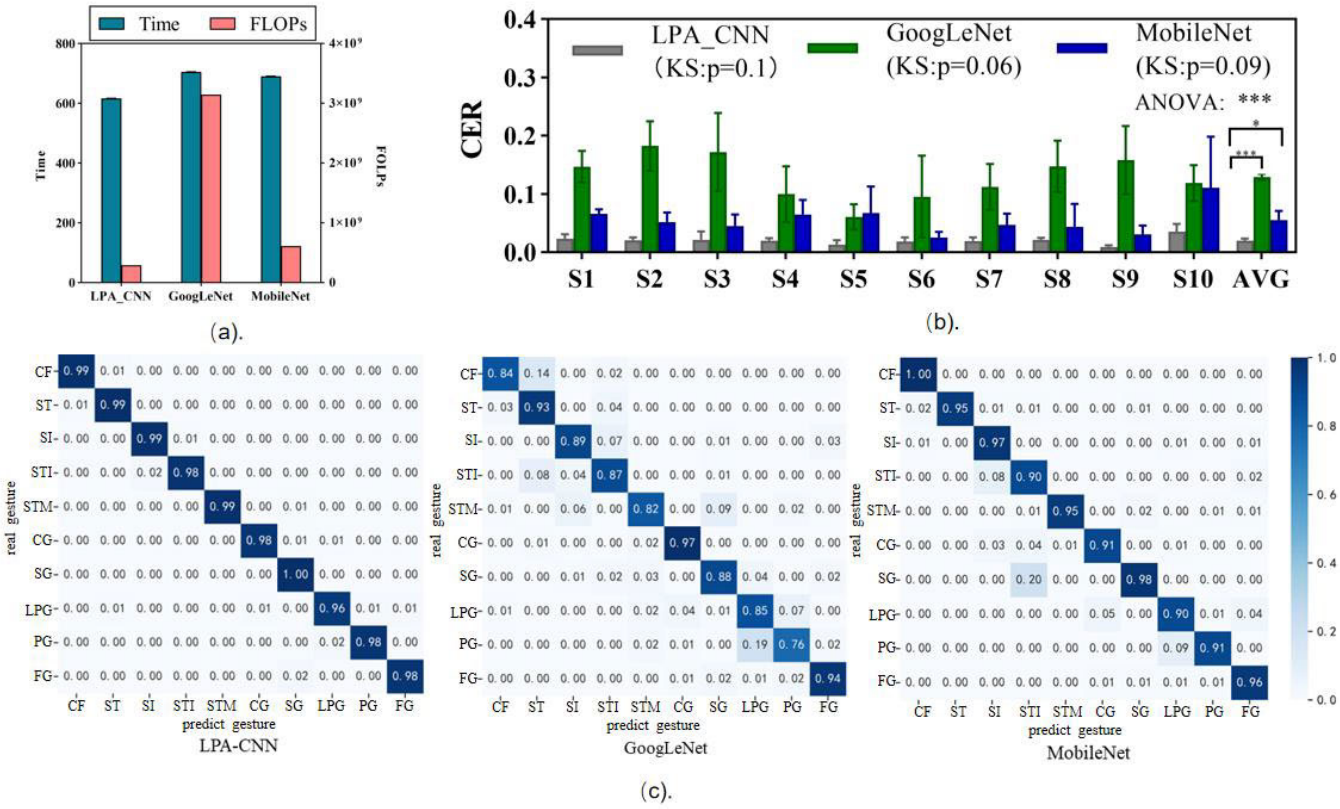


Fig. 13. LPA-CNN's recognition performance results. (a) Comparison of time and FLOPs across different models. (b) Gesture classification error rate of ten different subjects across three models, and * represents $P < 0.05$, *** represents $P < 0.001$. (c) Confusion matrix plots for LPA-CNN, GoogLeNet and MobileNet.

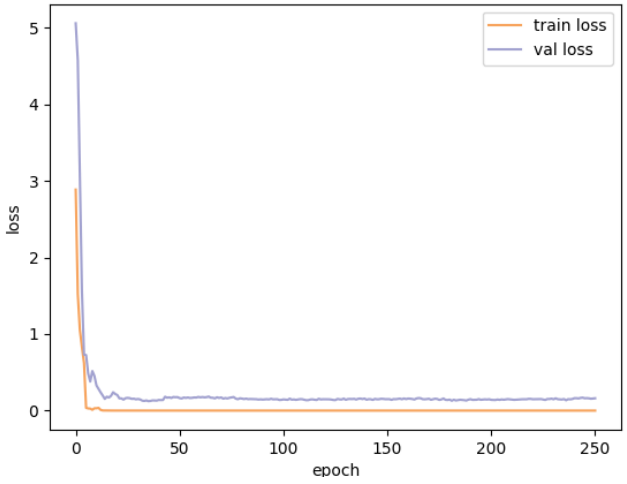


Fig. 14. Learning curve of the LPA-CNN Model.

and reduces computational cost, thereby improving overall efficiency. This architectural design offers new perspectives for future research on efficient neural networks.

VI. Conclusion and Future Work

This paper transforms the problem of HGR based on A-mode ultrasound signals into a classification problem using GAF encoding images and designs a lightweight model termed LPA-CNN with an average recognition accuracy of 0.98±0.02. First, we compared the impact of various feature extraction methods on the classification model through SC and RA evaluation metrics. We found that the ES method extracts more representative features, enabling the model to recognize gestures more effectively. Then, GAF encoding images exhibits higher sensitivity against the MTF and RP approaches in HGR through RA evaluation. Finally, we compared the developed LPA-CNN to other standard CNN, calculating the Time, FLOPs, and CER of each model. Experimental results show that the LPA-CNN model exhibits both computational efficiency and superior recognition accuracy.

A limitation of the proposed framework is that all experiments were conducted in offline conditions. Furthermore, this study primarily focused on HGR within specific individuals. In the future, an online HGR and the cross-subject generalization capabilities of the LPA-CNN model will be further investigated.

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